Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/19/2014
PI Name:	Miller, Fletcher J Ph.D.		
Project Title:	Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Diego	**State:**	CA
Zip Code:	92182-1323	**Congressional District:**	53
Comments:	Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2014 per NSSC information (Ed., 1/22/14) NOTE: End date is now 12/31/2013 per PI (Ed., 10/11/2012)		
Key Personnel Changes/Previous PI:	During the summer of 2010 Prof. George Sidebotham from Cooper Union participated in the research as a NASA Summer Faculty Fellow at NASA Glenn Research Center.		
COI Name (Institution):	Olson, Sandra (NASA Glenn Research Center) Wichman, Indrek (Michigan State University)		
Grant/Contract No.:	NNX10AD96A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research will study flame spread over flammable solid materials such as plastic, paper, cloth, and hair, using a new apparatus designed to simulate microgravity flow conditions on Earth. We designed and built two new Narrow Channel Apparatuses at San Diego State, and one at Michigan State, that control environmental variables such as flow velocity, gas pressure, and oxygen concentration to reproduce space craft atmospheres. Additionally, the Narrow Channel suppresses buoyant flow by having a closely spaced ceiling on the channel, so that flow conditions closely resemble those found on space craft. The objective of the research is to prove the Narrow Channel Apparatus as a viable test method for opposed flow material flammability testing. This will allow NASA to screen materials in more realistic conditions than does the current NASA test which makes no account of microgravity. The experimental research consists of test wherein we ignite various solids in the channels and measure how fast the flame spreads in various conditions. To accompany the experiments, we are developing a numerical model so that we can determine certain quantities such as heat loss to the channel ceiling that are difficult to measure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained from experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps.
Task Progress:	The SDSU and MSU Narrow Channel Apparatuses were used to recreate spacecraft atmospheric conditions and study the burning of plastic samples subject to various pressures, oxygen levels, and flow velocities. As shown previously, the Narrow Channel also reduces the effect of gravity and simulates microgravity conditions by eliminating the tendency of hot air to rise. The samples burned were 2 inches wide and up to 10 inches long, with thicknesses varying from .002 inches to 3/8”. A new filtering system was added to the back of the Narrow Channel Apparatus to remove particulate so that additional sensors could be added. These include CO and CO2 sensors to measure the product gases that are evolved in the simulated microgravity conditions. In addition, a new way to measure the surface temperature of the fuel material using an embedded thermocouple was developed. Numerical modeling of the flow and combustion process was conducted for the first time using the Fire Dynamic Simulator software. A two-dimensional domain was created to model the Narrow Channel. Two channels were modeled, one with a moving top (a Couette Channel) and one with the flow being driven by pressure differences. Both of these channels have been built and tested experimentally. Flame spread rate and temperature were calculated using the software for a variety of conditions that matched some of the experiments. The numerical model takes over a week to run in most cases, so that simulating many cases is time consuming. The experimental tests of the thin plastics in the Narrow Channel Apparatus showed good agreement with actual microgravity tests conducted in Japan and at NASA Glenn Research Center. Flame spread rates increased with oxygen level and with flow velocity up to about 15 cm/s. Both the 21% and 30% oxygen levels showed peaks, after which the flame spread rate fell as the flow velocity was turned up higher. The 50% oxygen flames kept spreading faster as the flow was increased. The agreement between the simulated and actual microgravity was very encouraging, indicating that this is a good test method for thin fuels. The effect of channel height was also explored. The data show that for gap heights above about 14 mm, the flame spread rate is not too dependent on gap height. However, other tests show that the gap needs to be about 10 mm or less or buoyancy starts to influence the results. For that reason, a gap height of 10 mm was used for these tests, even though the flame does experience some heat loss to the top of the channel. Experimental data from the thick fuel tests showed a strong variation with flow velocity, much more than for the thin fuels. A strong peak was reached about 20 cm/s. These data were corroborated by tests in the MSU Narrow Channel Apparatus as well. Tests there were also conducted with black and clear plastic to determine the difference. Limited numerical model results were obtained due to the time it takes for each run. However, for those tests that were completed, the numerical model shows the same trends with flow velocity, but the flame spread rate is much higher (three or more times) than has been measured experimentally. The reason for that discrepancy is under investigation.
Bibliography Type:	Description: (Last Updated: 06/15/2021)
Dissertations and Theses	Hung K. "Flame Spread Experiments in a Simulated Microgravity Environment Using a Couette Flow Apparatus." Master's Thesis, San Diego State University, Fall 2012. , Dec-2012
Dissertations and Theses	Hamdan G. "Numerical Modeling of a Planar Couette Flow Apparatus for Material Flammability Tests." Master's Thesis, San Diego State University, Fall 2013. , Dec-2013
Papers from Meeting Proceedings	Hung K, Miller F, Olson S. "Flame Spread Experiments in a Simulated Microgravity Flow Environment Using Laminar Planar Couette Flow." 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. AIAA paper 2012-3537. Paper AIAA-2012-3494. http://dx.doi.org/10.2514/6.2012-3494 , Jul-2012
Papers from Meeting Proceedings	Pepper J, Miller F, Olson S, Wichman I. "A Study of the Effectiveness of a Narrow Channel Apparatus in Simulating Microgravity Flame Spread over Thin Fuels." 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. Paper AIAA-2012-3493. http://dx.doi.org/10.2514/6.2012-3493 , Jul-2012

Papers from Meeting Proceedings	Bornand G, Miller F, Pepper J, Olson S, Wichman I. "Opposed-Flow Flame Spread in a Narrow Channel Apparatus over Thin PMMA Sheets." 8th US National Combustion Meeting, Park City, Utah, May 19-22, 2013. 8th US National Combustion Meeting 2013. Proceedings of a meeting held 19-22 May 2013, Park City, Utah, USA. Livermore, Calif. : Western States Section/Combustion Institute, 2013. p. 3740-3747. Paper #070FR-0418. Printed by Curran Associates, Inc. ISBN: 9781627488426 , May-2013
Papers from Meeting Proceedings	Hung KW, Miller FJ, Olson SL. "Flame Spread Experiments in a Simulated Microgravity Flow Environment Using Laminar Planar Couette Flow." Western States Section of the Combustion Institute Spring Technical Meeting 2012, Tempe, AZ, March 19-20, 2012. Western States Section of the Combustion Institute Spring Technical Meeting 2012. Livermore, Calif. : Western States Section/Combustion Institute, 2012. p. 302-317. Paper 12S-24. Printed by Curran Associates, Inc. ISBN: 978-1-62276-124-1. , Mar-2012
Papers from Meeting Proceedings	Pepper JM, Miller FJ, Olson SL. "Opposed Flow Thin Fuel Flame Spread in a Narrow Channel Apparatus." Western States Section of the Combustion Institute Spring Technical Meeting 2012, Tempe, AZ, March 19-20, 2012. Western States Section of the Combustion Institute Spring Technical Meeting 2012. Livermore, Calif. : Western States Section/Combustion Institute, 2012. p. 137-144. Paper 12S-10. Printed by Curran Associates, Inc. ISBN: 978-1-62276-124-1. , Mar-2012